

3 Species richness and an index of diversity can be used to measure biodiversity within a community.

0 3 . 1 What is the difference between these two measures of biodiversity?

[1 mark]

Scientists investigated the biodiversity of butterflies in a rainforest. Their investigation lasted several months.

The scientists set one canopy trap and one understorey trap at five sites.

- The canopy traps were set among the leaves of the trees 16–27 m above ground level.
- The understorey traps were set under trees at 1.0–1.5 m above ground level.

The scientists recorded the number of each species of butterfly caught in the traps. **Table 1** summarises their results.

Table 1

Species of butterfly	Mean number of butterflies		P value
	In canopy	In understorey	
<i>Prepona laertes</i>	15	0	< 0.001
<i>Archaeoprepona demophon</i>	14	37	< 0.001
<i>Zaretis itys</i>	25	11	> 0.05
<i>Memphis arachne</i>	89	23	< 0.001
<i>Memphis offa</i>	21	3	< 0.001
<i>Memphis xenocles</i>	32	8	< 0.001

0 3 . 2 The traps in the canopy were set at 16–27 m above ground level. Suggest why there was such great variation in the height of the traps.

[1 mark]

0 3 . 3

By how many times is the species diversity in the canopy greater than in the understorey? Show your working.

Use the following formula to calculate species diversity.

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

where N is the total number of organisms of all species and n is the total number of organisms of each species.

[3 marks]

Answer = _____

0 3 . 4

The scientists carried out a statistical test to see if the difference in the distribution of each species between the canopy and understorey was due to chance. The P values obtained are shown in **Table 1**.

Explain what the results of these statistical tests show.

[3 marks]

[Extra space]

